

Hospital Analytics

End-to-End Project Report

An end-to-end analytics pipeline on a synthetic US hospital EHR dataset, combining SQL Server data engineering, Power BI dashboarding, and Python machine learning to surface operational, financial, and clinical-risk insights across 27,891 patient encounters from 974 patients between 2011 and 2022.

Encounters analysed	Patients	Total claim cost	Model AUC
27,891	974	\$101.5M	0.907

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Tech stack	SQL Server · T-SQL · Power BI · Python · XGBoost · SHAP
Repository	github.com/jumma786/hospital-analytics-end-to-end
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Executive Summary

This report documents an end-to-end hospital analytics project covering data engineering, business intelligence, and machine learning. The work was conducted on synthetic EHR data (Synthea) but the methodology mirrors real production analytics: schema design, BULK INSERT loading, rigorous data quality auditing, multi-layered analysis, and explainable modelling.

Headline findings

Encounter volume. Volume peaked in 2014 (3,885) and 2021 (3,530). Ambulatory dominates every year except 2021, when outpatient surged to 40.2% — likely a COVID-era coding shift.

Payer coverage gap. 48.7% of encounters had zero payer coverage. 4,779 of those were insured patients (deductibles or denied claims), not just the uninsured — a nuance reporting commonly misses.

Cost concentration. Electrical cardioversion is the single largest cost driver: 1,383 procedures × ~\$25,903 ≈ \$35.8M. ICU admissions are rare (n=5) but cost \$206K each on average.

Payer disparities. Average claim cost varies 3.7× across payers, with Medicaid and uninsured patients carrying the highest costs (\$6,205 / \$5,593) — consistent with delayed-care patterns documented in US healthcare research.

Readmission prediction. An XGBoost model achieves AUC 0.907 predicting 30-day readmission, driven by clinical features (prior encounter count, recency, claim cost, age) rather than demographics.

Project phases

Phase	Deliverable	Status
Data engineering	Schema, BULK INSERT, data quality	Complete
SQL analytics	10 business-question queries	Complete
Power BI dashboard	3 pages: Operations, Financial, Patient Risk	Complete
Python ML model	30-day readmission, XGBoost + SHAP	Complete

1. Data Engineering & Quality

The project began with a synthetic EHR dataset from Synthea (Walonoski et al., JAMIA 2018) consisting of five CSV files: patients, encounters, procedures, payers, and organisations. Total volume: 974 patients, 27,891 encounters, 47,701 procedures, 10 payers, 1 organisation, spanning January 2011 to February 2022.

Loading issues discovered and resolved

Issue 1 — Missing patient row. SQL Server's BULK INSERT failed silently on one patient record (Mrs. Melaine933 Hintz995, 19 dependent encounters) due to special characters in a quoted field. Detected via LEFT JOIN orphan check. Resolved using SSMS's Import Flat File Wizard which handles quoted CSV fields more robustly than BULK INSERT.

Issue 2 — Empty strings instead of NULLs. All optional fields (DEATHDATE, ZIP, MARITAL) loaded as empty strings rather than NULL. This made every patient appear alive in queries — a silent failure that would have invalidated every mortality-related metric downstream. Resolved with `NULLIF()` conversions and a TRY_CAST reload that produced 154 patients with valid death dates and 142 with missing ZIPs — exactly matching the source CSV.

Findings documented but not 'fixed'

Finding	Interpretation	Decision
70% null reasoncodes in encounters	Routine visits have no diagnosis — heavily skewed	leave as is NEHR
53 encounters dated after patient's death	likely admin/billing entries posted post-mortem	keep for Keep for most analysis; exclude from patient behaviour
One patient with 1,381 lifetime encounters	chronic care profile (likely dialysis)	Flag in readmission analysis to prevent misinterpretation
4,779 zero-coverage encounters	Disputed or denied claims, not unreported	Report both interpretations explicitly

Documenting these decisions explicitly — rather than silently filtering or imputing — is the difference between query writing and analysis. Each call is reviewable, and stakeholders can disagree with the interpretation without invalidating the underlying data.

2. SQL Analytics

Ten business questions were posed across three objectives. All queries were written in T-SQL and executed against the local SQL Server instance. Below is a summary of each finding; full SQL is in the repository's `/sql/` folder.

Objective 1: Encounters Overview

Question	Answer
Total encounters per year	Range 220 (2022, partial) to 3,885 (2014). Peaks in 2014 and 2021.
Class mix per year (% share)	Ambulatory 37–60% every year. 2021 anomaly: outpatient surged to 40.17%.
Over vs under 24 hours	Under 24h: 26,739 (95.87%). Over 24h: 1,152 (4.13%).

Objective 2: Cost & Coverage Insights

Question	Answer
Encounters with zero payer coverage	13,586 (48.71% of all). See nuance below.
Top 10 most frequent procedures	Routine assessments (Synthea \$431 flat rate). Renal dialysis (#6) is the cost outlier at \$1,000.
Top 10 highest-cost procedures	ICU admit (\$206K, n=5), CABG (\$47K, n=9), Hemodialysis (\$29K, n=27).
Avg claim cost by payer	Medicaid \$6,205 / NO_INSURANCE \$5,593 / Medicare \$2,167. 3.7x range.

Key analytical nuance. 'Zero payer coverage' could mean either `PAYER_COVERAGE = 0` or `PAYER = NO_INSURANCE` — and the two return different numbers (13,586 vs 8,807). The gap is 4,779 insured patients whose claims were nonetheless covered \$0 — consistent with high-deductible plans or denied claims. Both numbers are reported.

Objective 3: Patient Behaviour Analysis

Question	Answer
Unique patients per quarter	45 quarters. Stable ~240/qtr most of the period. Surges in 2014-Q1 (394) and 2021-Q1/Q2 (377).
Patients readmitted within 30 days (broad)	772 of 974 patients (79.3%)
Patients readmitted (strict, inpatient-only)	29 patients (CMS-aligned definition)
Top readmitter	Kimberly627 Collier206 — 1,375 readmissions out of 1,381 lifetime encounters (chronic ca

3. Power BI Dashboard

A three-page Power BI Desktop dashboard connected to the SQL Server instance via Import mode. Built on a star schema with *encounters* as the fact table, surrounded by four dimensions (patients, payers, organisations, procedures). All measures are DAX-defined; calculated columns are used only for the encounter-duration bucket.

Page 1: Operations Overview

Four KPI cards (total encounters, patients, total claim cost, 30-day readmission rate) over a year-over-year trend line, an encounter-class donut, and a class-mix-over-time stacked bar. The 2014 and 2021 volume peaks read at a glance.

Page 2: Financial Overview

Surfaces the most analytically valuable finding: the coverage gap. Includes an average-claim-cost-by-payer bar chart, a coverage-type donut (insured vs uninsured), a top-10-procedures-by-cost bar, and a treemap of financial burden by encounter class.

Page 3: Patient Risk & Readmission

Clinical-readmission rate (strict, inpatient-only definition), readmitted-patients count, mortality rate, and average stay duration. Risk distribution by age group and gender. Mortality breakdown by gender and encounter class.

The .pbix file is embedded with cached data, so visuals render without a live SQL Server connection. Recipients can open and view the dashboard without needing the source database.

4. Readmission Prediction Model

A machine learning pipeline predicting whether an encounter leads to another encounter within 30 days. Implemented in Python with Scikit-learn, XGBoost, and SHAP. The notebook connects directly to SQL Server via SQLAlchemy and pyodbc — features are engineered in SQL (window functions in a single CTE), and the resulting feature matrix is loaded into Pandas.

Approach

- **Feature engineering in SQL** — sequential features (prior encounter count, days since previous visit) computed using LAG, LEAD, and COUNT OVER window functions. Avoids expensive Python loops.
- **Two models compared** — Logistic Regression as an interpretable baseline; XGBoost as the stronger non-linear model. If the baseline already performs well, additional model complexity is not justified.
- **Train/test split** — 80/20 stratified on the target variable. 22,312 train / 5,579 test, both with identical 61.6% / 38.4% target distribution.
- **Explainability** — SHAP values for the XGBoost model on 500 test predictions. Both summary bar plot and beeswarm plot generated.

Performance

Model	AUC	Notes
Logistic Regression	0.800	Interpretable baseline. Strong signal already detected.
XGBoost	0.907	10.7-point AUC gap justifies the additional complexity.

Context. Real-world readmission models on production EHR data typically achieve AUC 0.65–0.75. The higher score here reflects Synthea's cleaner patterns relative to real EHR noise, not a real-world performance claim. The methodology, however, transfers directly.

Feature importance (SHAP)

The model's top predictors are clinical and behavioural, not demographic — exactly what a credible healthcare model should look like:

Rank	Feature	Direction of effect
1	prior_encounter_count	More prior encounters → higher readmission probability
2	days_since_prev	Recent prior visit → higher readmission probability
3	claim_cost	Higher cost (sicker patient) → higher readmission probability
4	patient_age	Older patient → modestly higher readmission probability
5	duration_hours	Longer encounter (sicker patient) → higher readmission probability
6	encounter_class_wellness	Wellness check → pushes AWAY from readmission

Race and gender appear near the bottom of the importance ranking — the right outcome for an ethically credible clinical model.

Limitations

- **Synthetic data.** Synthea generates patient histories from population statistics, not real clinical pathways. Real-world data is noisier and the gap will narrow.
- **Broad target definition.** Any encounter within 30 days counts. The CMS-aligned strict definition (inpatient-to-inpatient) would yield a separate, harder modelling problem worth pursuing.
- **Chronic-care outlier.** One patient with 1,381 encounters heavily influences the model. A production version would add a chronic-care stratification feature.

5. Conclusions & Recommendations

Business takeaways

- **Coverage gap is wider than 'uninsured' alone.** 35% of zero-coverage encounters belonged to insured patients. Hospital finance teams should track both metrics; reporting only 'uninsured rate' under-counts the patients exposed to large out-of-pocket costs.
- **Cost is concentrated, not diffuse.** One procedure type (electrical cardioversion) drives ~\$35M of total claim cost. Operations teams can prioritise cost-management efforts on a small number of high-impact procedures rather than spreading thinly across all encounter types.
- **Medicaid and uninsured patients carry the highest average costs.** Consistent with US healthcare research on delayed-care patterns. Targeted preventative outreach could reduce long-term cost more effectively than acute-care interventions alone.
- **Readmission risk is predictable from clinical history.** Prior visit count and recency dominate the model. Hospitals could integrate a simplified version of the model into discharge planning to flag high-risk patients for follow-up.

Technical takeaways

- **BULK INSERT requires verification, not faith.** Two silent failures (one missing row, every empty string becoming " instead of NULL) would have invalidated downstream analytics. Always sanity-check after a bulk load.
- **Documenting interpretation matters more than computing.** Three of the ten business questions admitted multiple defensible interpretations. Surfacing those decisions explicitly is what separates analysis from query writing.
- **Explainable models build trust.** SHAP showed that the XGBoost model relied on clinically meaningful features, not race or gender. A 0.95-AUC model that secretly relies on a demographic proxy is far more dangerous than a 0.85-AUC model that's transparent about its reasoning.

Next steps

- Add a chronic-care stratification feature and re-evaluate model performance.
- Calibrate predicted probabilities (Platt scaling or isotonic regression) for use in clinical decision-support.
- Replicate the methodology on real EHR data via a partnership with an NHS trust or US health system.
- Wrap the model in a REST API so it could be called from a discharge-planning UI.

Repository: github.com/jumma786/hospital-analytics-end-to-end

Data source: Walonoski J, Kramer M, Nichols J, et al. Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record. JAMIA, March 2018.